

RangeSure

An AI-native trip-decision layer for Tata Motors Commercial Vehicles.

DTAI Capstone 2026 · IIM Lucknow | Live app: rangesure.vercel.app

WHAT IT DOES

Before a truck is dispatched, RangeSure predicts its real-world range, energy cost and charging need for a specific route — and states whether the trip can be trusted. It lets electric fleets run at full utilisation instead of planning defensively around uncertain range.

THE AI INSIDE

A per-segment **physics model** predicts energy use; a **calibration model** learned on 1,500 trips corrects it; **Gemini** turns the numbers into plain-language guidance and never computes one itself. A governance **red-line** withholds a confident verdict when data is missing or the margin is thin.

DATA SOURCES

OpenRouteService (routing, elevation, geocoding · driving-hgv), Open-Meteo (weather, 4 points/route), Open Charge Map (charging), Google Gemini 2.5-flash. All keys are server-side. The prototype runs on public data + published Tata specs; the pilot adds live BMS / Fleet Edge data.

The app, tab by tab — what you put in, what you get out, and the decision it drives

1

Plan Trip — Driver / Dispatcher

INPUTS

Vehicle (Ace EV · Ultra E.9 · Prima E.28K · Prima E.55S) · origin · destination · payload (kg, capped at the vehicle's max) · current battery % · electricity tariff (₹/kWh) · delivery window (optional) · one-click demo presets.

OUTPUTS

A clear verdict — **complete** · **charge once** · **needs N stops** · **not feasible (manual planning advised)** — plus distance, energy use (kWh/km, physics → calibrated), predicted full range, energy cost ₹/km, arrival battery %, charging stop with detour and added minutes, a best/expected/worst repeatability band, confidence range, a map with the steepest climb named, and Gemini driver guidance + a free-form "ask" box.

HOW TO USE IT

Answers the one question that matters before dispatch: **will this truck complete this trip safely, and at what cost per km?** Loads are assigned against a known arrival buffer instead of a driver's guess, so trucks run closer to full capacity; the red-line prevents the single confident-but-wrong call that strands a vehicle and breaks a delivery window. [cost-per-km](#) [range confidence](#)

2

Guarantee — Commercial / Sales

INPUTS

The trip computed in Plan Trip · a completion-guarantee level (90–99.5% slider) · pricing assumptions from Settings (margin, disruption cost).

OUTPUTS

A committed **₹/km price** for the chosen guarantee level · a cost breakdown (energy + risk buffer + margin) · a moat chart showing the risk buffer shrinking as corridor trips grow, with "0 trips = cannot underwrite".

HOW TO USE IT

Turns a prediction into a **sellable commitment** — the shift from selling certified range to selling predictable trip economics and cost-per-km contracts. Sales can quote a price and a completion guarantee per corridor; the buffer curve shows why only Tata's 1M+ vehicle Fleet Edge data can underwrite it, and why both price and moat improve with every trip. [recurring revenue](#) [EV conversion](#)

3

Fleet — Fleet Manager

INPUTS

A set of trucks/trips (route, payload, battery) · per-row vehicle re-assignment · a house guarantee level.

OUTPUTS

A sortable table (distance, kWh/km, ₹/km, range, arrival %, feasibility) · a dashboard strip (fleet total energy, average ₹/km, count needing a charge stop) · marginal/infeasible flags · a guarantee-book roll-up of total committed contract value.

HOW TO USE IT

Moves single-trip planning up to **portfolio decisions** — which routes are marginal, which truck to assign where, and which corridors are underwritable today versus need more data before quoting. [dispatch](#) [fleet uptime](#)

4

Charging — Driver / Ops

INPUTS

A location or city to search.

OUTPUTS

Nearby charging stations plotted on a map, each listed with its distance (km) from the searched point, sorted nearest-first.

HOW TO USE IT

Closes the loop from "a stop is needed" to "**here is the reachable one**" — confirming charging options around a hub or planned stop and cutting unplanned-stop risk. [fleet uptime](#)

5

Reports — Management

INPUTS

Automatically aggregated from every trip logged in Plan Trip (no manual entry).

OUTPUTS

Trips logged · total distance · total energy (kWh) · total & average cost (₹/km) · number needing a charge stop · % **feasible without a stop**.

HOW TO USE IT

The **evidence layer** for the pilot: it quantifies the problem in the fleet's own data (in testing, 0% of real trips were feasible without a stop), tracks whether ₹/km improves over time, and feeds the KPI gates — range error $\leq 10\%$, energy improvement $\geq 6\%$, unplanned stops down $\geq 30\%$. [pilot KPIs](#)

6

Settings — Admin / EV Engineer

INPUTS

Editable vehicle physics parameters (empty mass, Crr, Cd, frontal area, drivetrain & regen efficiency, aux load, battery) and pricing (tariff, charger power).

OUTPUTS

Changes apply immediately to every Plan Trip and Fleet calculation and are saved in the browser.

HOW TO USE IT

Keeps the model **transparent and tunable** — every assumption is visible and owned (the EV engineer calibrates parameters to a real vehicle variant), which is what makes the numbers defensible to a board or an audit rather than a black box. [governance](#)

KNOWN LIMITATIONS — STATED HONESTLY

Physics parameters are engineering estimates until calibrated against live BMS data; the flagship Mumbai–Pune corridor runs near its feasibility margin, so a specific verdict can shift with live weather at request time; calibration is trained on synthetic-but-realistic trip history standing in for Fleet Edge telematics. These are exactly the items the proposed 12-month pilot is designed to validate.

Figures reflect a live app run on 23 Aug 2026 — Ultra E.9, Mumbai → Pune: 165 km · 1.09 kWh/km · ₹9.29/km · 183 km predicted full range.